

Emerging Trends in the Design of Cognitive IoT

This is an extract from a speech presented by Dr Manjunath Iyer, principal consultant, Wipro, at iotshow.in in 2019. It describes the emerging trends in the design of cognitive Internet of Things (IoT). Rather than just capturing data, cognitive IoT devices can think, understand and accordingly provide data to the user.



Things were there even before the term Internet of Things (IoT) was coined, but these were not connected to the Internet. Today, with the introduction of the Internet, everyday objects (things) have become smart and capable enough to take decisions on their own.

The next step is transformation of these IoT devices into cognitive IoT devices. So, rather than just capturing the data, these devices should also have the ability to think, understand, and accordingly provide data to the user. IoT devices can be turned into cognitive IoT devices by incorporating cognitive computing technologies.

Living IoT devices

Yes, even an animal like buffalo or a cow can be transformed into an IoT object. How? Just connect a location sensor to the ear of the animal. Based on the animal's roaming information, the user can know grazing pattern of the animal on a particular day, how much milk it will provide, and thus calculate the amount of profit it will generate.

A living being as small as a honeybee can also be transformed into an IoT object. A small nanosensor can be placed on the body of the insect and, based on its flight path, one can deduce what kind of flowers that honeybee is interested in. This can lead to planting more of such flowers and production of greater amount of honey, resulting in more profit generation.

IoT is not free from issues, however. You cannot put anything and everything, anywhere and everywhere; there are a lot of connectivity issues that you need to deal with.

Let us first understand what an IoT system consists of. It mainly has an object or thing to be connected, connectivity through wireless standards, and the Internet.

The object (things, organisms, etc) to be connected is the main component of the entire IoT system. Connectivity comes in various forms such as Bluetooth, Wi-Fi, Zigbee, NFC, RFID, and many other wireless standards. The data is put on the Internet for storage, processing, and analytics.

When 5G becomes operational, it will be easier to connect many more

devices and put massive amount of data into a network.

IoT is a culmination of such technologies as augmented reality, signal processing, Big Data, mobile computing, object and voice detection, and many more. It can be used in various sectors like banking, energy, industrial, vending, security and surveillance, medical, consumer, telematics, etc. Due to such huge diversity, the chances of it going wrong increases. Therefore the right combination of components should be selected keeping in mind the end result and its applications.

Example of IoT based solution

Consider a mobile phone that remotely operates the lights at home.

This can happen by first registering the objects to be monitored (home lights in this case) on an app installed on the phone. Next, by object identification. Each and every object (light) can be subscribed to the installed application running on the Internet. Through this, one can easily control all the lights.

